

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester IV Mid-Term Examination, 2025-26 (NEP)

DEPARTMENT OF MATHEMATICS**Paper Code: MATMJ-41 : Calculus of Several Variables (Mid-Term)**

Time: 1 Hour

Maximum Marks: 20

Instructions: Attempt any FIVE questions. All questions carry equal marks (4 marks each).

Question 1. Define double limit and repeated limits for $f(x, y)$ at $(0, 0)$. Show that for $f(x, y) = \frac{xy}{x^2 + y^2}$, repeated limits exist and are equal, but double limit does not exist. [4]

Question 2. State and prove Schwarz's Theorem on the equality of mixed partial derivatives $f_{xy}(a, b)$ and $f_{yx}(a, b)$. [4]

Question 3. State and prove Euler's Theorem on homogeneous functions of two variables x and y of degree n . [4]

Question 4. Discuss the maxima and minima of $f(x, y) = x^3 + y^3 - 3axy$. [4]

Question 5. State Lagrange's Method of Undetermined Multipliers and find the extreme values of $f(x, y, z) = x^2 + y^2 + z^2$ subject to $x + y + z = 3$. [4]

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